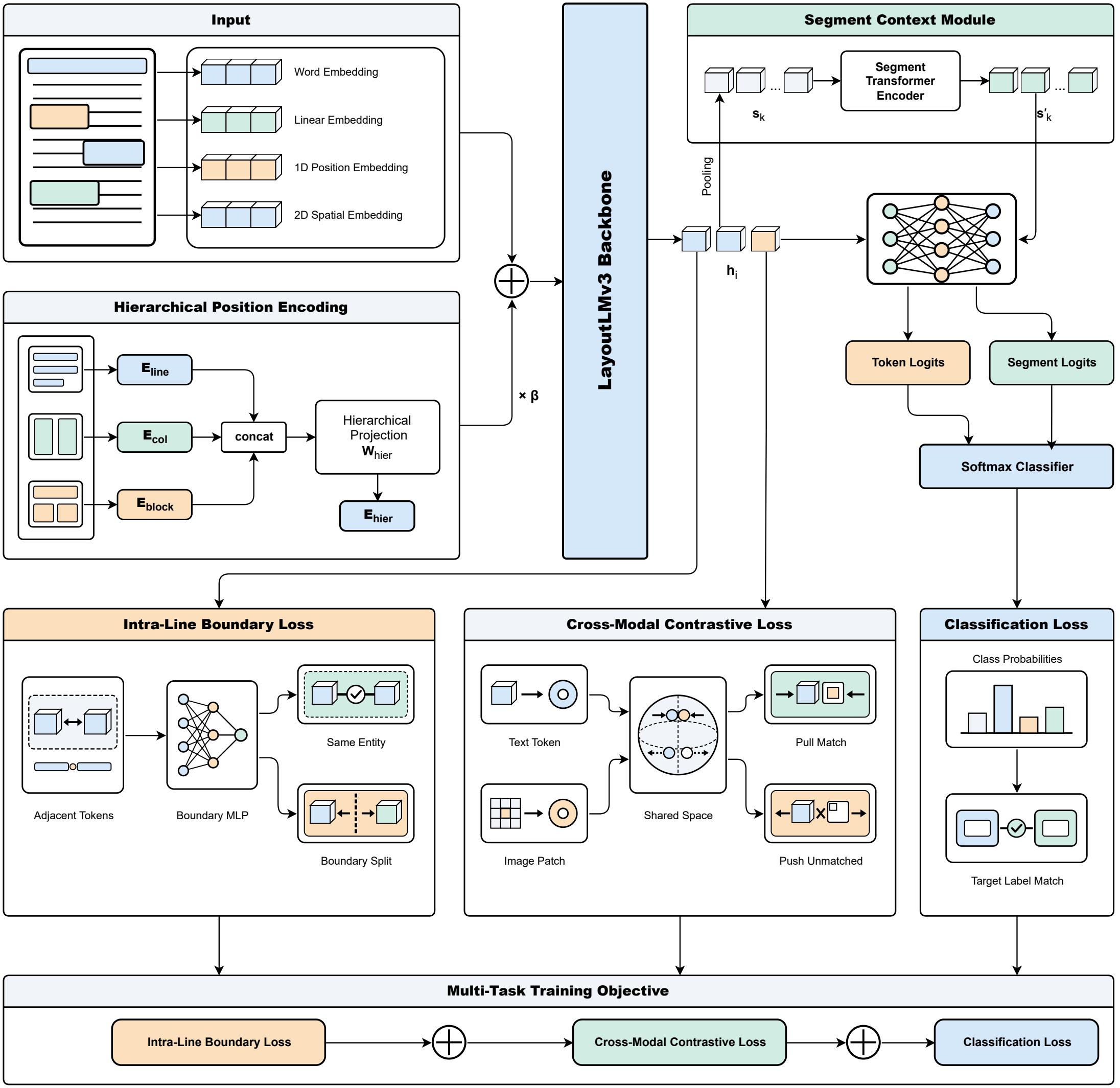

Input
Word Embedding
Linear Embedding
1D Position Embedding
2D Spatial Embedding
Hierarchical Position Encoding
E line
E col
E block
concat
Hierarchical Projection W hier
E hier
× β
LayoutLMv3 Backbone
Segment Context Module
Segment Transformer Encoder
s k
s′ k
Pooling
h i
Token Logits
Segment Logits
Softmax Classifier
Intra-Line Boundary Loss
Adjacent Tokens
Boundary MLP
Same Entity
Boundary Split
Cross-Modal Contrastive Loss
Text Token
Image Patch
Shared Space
Pull Match
Push Unmatched
Classification Loss
Class Probabilities
Target Label Match
Multi-Task Training Objective
Intra-Line Boundary Loss
Cross-Modal Contrastive Loss
Classification Loss